

COMSATS UNIVERSITY ISLAMABAD, LAHORE CAMPUS



PROGRAMMING FUNDAMENTALS
ASSIGNMENT NO.03
STUDENT MARKS ANALYZER
2-D ARRAYS

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SP26-BAI-019

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SEMESTER 01 (BAI-A)

STUDENT MARKS ANALYZER

CODE

```
#include <iostream>

using namespace std;

const int MAX=100;

//User-defined Function Prototypes

int TOTAL(int marks[][MAX], int row, int m);

void subject(int marks[][MAX], int n, int m);

void highLow(int marks[][MAX], int n, int m);

int main()
{
    int n, m;

    int marks[MAX][MAX];

    cout << "Enter total students= ";

    cin >> n;

    cout << "Enter total subjects= ";

    cin >> m;

    //Taking marks from the user

    for(int i=0; i<n; i++)
    {
        cout << "\nStudent " << i+1 << " marks= \n";

        for(int j=0; j<m; j++)
        {
            cout << " Subject " << j+1 << "= ";

            cin >> marks[i][j];

        }
    }
}
```

```

//Displaying marks
cout << "\n-- Marks Table --\n";

for(int i=0; i<n; i++)
{
    cout << "Student " << i+1 << ": ";
    for(int j=0; j<m; j++)
        cout << marks[i][j] << " ";
    cout << endl;
}

//Every Student's Total Marks
cout << "\n-- Student Report --\n";
for(int i=0; i<n; i++)
{
    int total = TOTAL(marks, i, m);
    cout << "Student " << i+1 << " | Total= " << total << " | Average= " << total/m << endl;
}

subject(marks, n, m);
highLow(marks, n, m);
}

//Function Definitions
int TOTAL(int marks[][MAX], int row, int m)
{
    int total=0;
    for(int j=0; j<m; j++)
        total += marks[row][j];
    return total;
}

void subject(int marks[][MAX], int n, int m)
{

```

```

cout << "\n-- Subject-Wise Report --\n";
for(int j=0; j<m; j++)
{
    int sum=0;
    for(int i=0; i<n; i++)
        sum += marks[i][j];
    cout << "Subject " << j+1 << " | Total= " << sum << " | Average= " << sum/n << endl;
}
}

void highLow(int marks[][MAX], int n, int m)
{
    int high=marks[0][0], low=marks[0][0];
    for(int i=0; i<n; i++)
    {
        for(int j=0; j<m; j++)
        {
            if(marks[i][j] > high)
                high=marks[i][j];
            if(marks[i][j] < low)
                low=marks[i][j];
        }
        cout << "\nHighest= " << high << " | Lowest= " << low << endl;
    }
}

```

OUTPUT

```
C:\Users\DO Star Asia\Documents\FARYAL's 1st Semester\PF\SP26-BAI-019 2D-Arrays.exe
Enter total students= 3
Enter total subjects= 3

Student 1 marks=
  Subject 1= 56
  Subject 2= 66
  Subject 3= 77

Student 2 marks=
  Subject 1= 98
  Subject 2= 45
  Subject 3= 66

Student 3 marks=
  Subject 1= 45
  Subject 2= 76
  Subject 3= 98

-- Marks Table --
Student 1: 56 66 77
Student 2: 98 45 66
Student 3: 45 76 98

-- Student Report --
Student 1 | Total= 199 | Average= 66
Student 2 | Total= 209 | Average= 69
Student 3 | Total= 219 | Average= 73

-- Subject-Wise Report --
Subject 1 | Total= 199 | Average= 66
Subject 2 | Total= 187 | Average= 62
Subject 3 | Total= 241 | Average= 80

Highest= 77 | Lowest= 56

Highest= 98 | Lowest= 45

Highest= 98 | Lowest= 45

-----
Process exited after 26.5 seconds with return value 0
Press any key to continue . . .
```